

A Parking Occupancy Detection Algorithm Based on AMR Sensor

Zusheng Zhang, Ming Tao, and Huaqiang Yuan

Abstract—Recently, with the explosive increase of automobiles in cities, parking problems are serious and even worsen in many cities. This paper proposes a new algorithm for parking occupancy detection based on the use of anisotropic magnetoresistive sensors. Parking occupancy detection is abstracted as binary pattern recognition problem. According to the status of the parking space, the recognition result contains two categories: 1) vacant and 2) occupied. The feature extraction method of the parking magnetic signal is proposed. In addition, the classification criteria are derived based on the distance discriminate analysis method. Eighty-two sensor nodes are deployed on the roadside parking spaces. By running the system for six months, we observed that the accuracy rate of the proposed parking occupancy detection algorithm is better than 98%.

Index Terms—AMR sensor, wireless sensor networks, parking occupancy detection.

I. INTRODUCTION

AT PRESENT, the sensors used in the traffic information acquisition include the following types: inductive loop detector [1], image (camera) sensor [2], [3], acoustic sensor [4], [5], infrared sensor [6], ultrasonic sensor [7], etc. The image sensor acquires an abundance of information, but it is vulnerable to bad weather and nighttime operation. The acoustic sensor and infrared sensor are vulnerable to noise in deployed environments. Magnetic sensors based on magneto resistors have recently been proposed for vehicle detection [8]–[11] because they are quite sensitive, small and more immune to environmental factors such as rain, wind, snow or fog than sensing systems based on video cameras, ultrasound or infrared radiation.

A. Dynamic Traffic Information Acquisition

Many algorithms have been proposed for dynamic traffic information acquisition system based on AMR sensor. The PATH program of the University of California,

Berkeley [12], [13] had first extensively explored of AMR sensor network based vehicle detection system. S. Y. Cheung et al. [13] had explored the applications for vehicle detection, speed estimation, and classification. Experiment results show that the vehicle detection accuracy rate is more than 99%, and the accuracy rate to estimate length and speed of vehicle is more than 90%.

M. Kang et al. [14] proposed a vehicle detector with an AMR sensor and addresses experimental study carried out to show the detector's characteristics and performance. Li Cui et al. [15], [16] proposed a Similarity Based Vehicle Detection (SBVD) algorithm to detect vehicles in low SNR conditions by calculating the correlation between on-road signals and a referential signal. Besides, data fusion algorithm based on fuzzy logic theory has also been proposed to monitor parking space using magnetic sensor [17].

S. Taghvaeeyan et al. [18] presents a method to estimate the position of the vehicles. It utilizes the fact that the magnetic moment caused by a vehicle decreases as the distance to the sensor increases. The proposed method is affected by the type and size of the vehicle. The author integrated two AMR sensors and a sonar sensor together to address this issue and improve the estimation.

B. Static Traffic Information Acquisition

Technologies normally employed within indoor car parks, such as ultrasonic sensors [19], could not be used outdoor because every space would need to be wired. Installing cables and wires along the roadside for data transmission and energy supply is very difficult and challenging. So a completely wireless solution was adopted. Wireless AMR sensor networks [20] have lots of potential toward providing an ideal solution for roadside parking service, such as their low power, small size, low cost, and high accuracy.

The use of AMR sensors for detection of vehicle in parking space has been previously studied. In [21], S. Lee discussed the effect of detecting vehicles by comparing the acoustic, visual light, infrared, temperature, ultrasonic, and AMR sensors. Their experiments verified that ultrasonic and AMR sensors have better accuracy and reliability in the parking space.

Threshold detection algorithms are usually to measure the geomagnetic field change of z-axis [22], [23]. If this change in frequency exceeds a threshold, the sensor will indicate the detection of a vehicle. But a single threshold cannot apply to all situations and often leads to a high false alarm rate. In order to solve this problem, an adaptive threshold

Manuscript received September 7, 2014; revised October 2, 2014; accepted October 2, 2014. Date of publication October 9, 2014; date of current version December 3, 2014. This work was supported in part by the Guangdong Province Natural Science Fund, under Grant S2013040016582, in part by the National Natural Science Foundation of China under Project 61402105 and Project 61300198, and in part by the Guangdong Higher School Scientific Innovation Project, China, under Grant 2013KJCX0177. The associate editor coordinating the review of this paper and approving it for publication was Prof. Yu-Cheng Lin.

The authors are with the Department of Computer Science and Technology, Songshan Lake Science and Technology Industry Park, Dongguan University of Technology, Dongguan 523808, China (e-mail: zushengzhang@gmail.com; taoming6723@126.com; hyuan66@163.com).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/JSEN.2014.2362122

algorithm is presented [24]. However, threshold-based vehicle detection algorithms have congenital defects. Firstly, there is no existence of a reasonable threshold to distinguish between the signal of interference and vehicle parking, especially in low Signal-Noise-Ratio (SNR). Furthermore, different places have different geomagnetic fields, so it is difficult to have a threshold to adapt all situations. Our simulations and experiments also show that the aforementioned algorithm is difficult to apply to the real system, because its vehicle detection success rate is less than 96%.

Another detection method is to study the characteristics of the wave pattern. J. Ding proposed Min-max Detection Algorithm [25]. Modified Min-max Algorithm for magnetometers [26] has also been published. These algorithms analyze the process of parking signal changes, but they still rely on single threshold and have low vehicle detection success rate.

Commercial sensors, such as SENSIT system [27], can detect parking occupancy. Each node was equipped with two sensors: infrared and magnetic, and its vehicle detection accuracy rate is nearly 100%. But the system is too expensive.

Existing parking occupancy detection algorithms did not analysis the characteristic between the parking signal and interference signal. On the other hand, a whole parking period of a vehicle can be divided into three phases: entering, parking stop and leaving. Most existing algorithms only consider the signal's characteristic of the parking stop phase. They did not analysis the relationship between the entering and leaving phases of magnetic signal. This paper will address these problems and propose a more accurate detection algorithm. Our main contributions are given as follows:

- By Analysis of the characteristics of the disturbance geomagnetic signal, the parking vehicle detection algorithm is proposed, and the algorithm has an immune feature for the interference signal.
- Vehicle leaving can be regarded as an inverse process of its entering. Choosing the signal of the entering phase as the referential signal, leaving detection is to computing the similarity between the two phases.

The remainder of this paper is organized as follows. Section II analyzes the characterizations of parking signal and proposes the algorithm. Section III conducts experiments to prove the performance of the algorithm. Finally, Section IV makes a brief conclusion.

II. PARKING OCCUPANCY DETECTION ALGORITHM

According to the status of the parking space, parking occupancy detection is a binary recognition problem (two cases: vacant and occupied). The key issue is to train the classification criteria from the magnetic signal. Solve the problem involve following steps: Firstly, analysis characteristics of the magnetic signal and computing the features variable of fluctuate signal. Secondly, classification rules are trained using the distance discriminate method, and the training data are pre-classified examples (class label is known for each example). Finally, the parking detection algorithm is designed based on classification criteria.

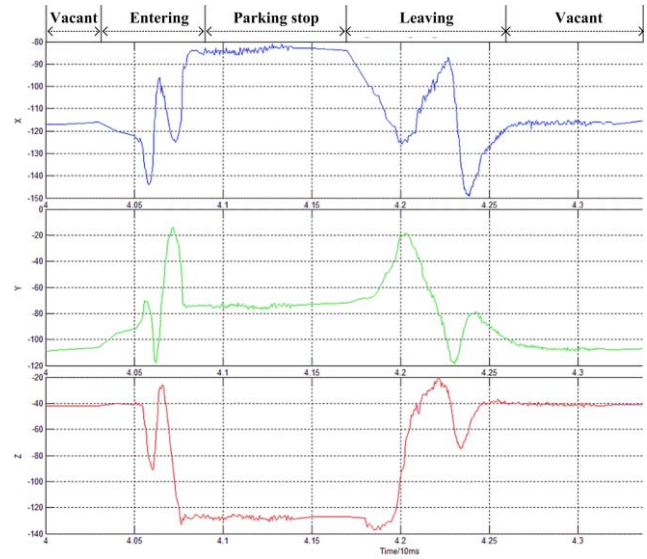


Fig. 1. Three-axes magnetic signature of a parking space.

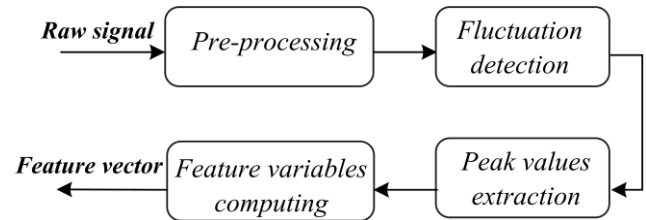


Fig. 2. Signal processing.

A. Feature Extraction of the Magnetic Signal

Sensor node consists of a 3-axis AMR sensor. Fig. 1 is the three-axis magnetic signature of a vehicle parking process. A whole parking period includes three phases: entering, parking stop and leaving. Initially, the parking space is vacant, and the values of x , y , z are the background's magnetic fields. Then a vehicle enters the parking space and creates a fluctuation of the magnetic field. After the vehicle parking stop, the vehicle creates a stable disturbance on the environment's magnetic field. The parking space is occupied. Then the vehicle is leaving the parking space, and also has a fluctuation signature. Finally, the parking space is vacant.

1) *Signal Preprocessing*: The geomagnetic signal is studied and summarized in the time domain involves a series of steps as shown in Fig. 2. For vehicle presence applications, the direction of the magnetic field is not important, but the detection of a significant shift and frequent fluctuation in the magnetic field are the key factors. The vector magnitude shift from the environment's magnetic field would be the most reliable method. Using digitized measurements of three-axis sensor outputs after amplification, the vector magnitude would be:

$$G(i) = \sqrt{X_i^2 + Y_i^2 + Z_i^2} \quad (1)$$

A smoothing filter, which takes a running average of the signal, is used to smooth the signal. The running average is

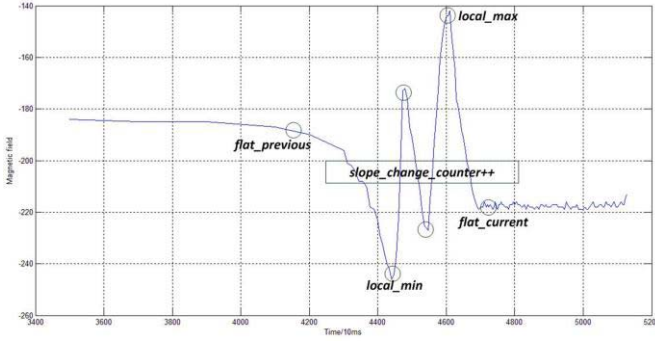


Fig. 3. Peak values of magnetic signal.

given by

$$A(i) = \begin{cases} \frac{G(i) + G(i-1) + \dots + G(1)}{L} & \text{for } i < L \\ \frac{G(i) + G(i-1) + \dots + G(i-L+1)}{L} & \text{for } i \geq L \end{cases} \quad (2)$$

$G(i)$ is the vector magnitude and L is the pre-defined running average buffer size. For implementation, the buffer G_{buf} stores the values of $[G(i), G(i-1), \dots, G(i-L+1)]$.

2) *Fluctuation Detection*: Since the fluctuation of the magnetic signal is a key characteristic of the vehicle entering and leaving. Exact detection the fluctuation signature is important for the vehicle detection algorithm. $F(i)$ is the stable states of the magnetic signal. If there has one up-or-down fluctuation sample of the magnetic signal in the G_{buf} , the signal is unstable and $F(i) = 1$, otherwise, the signal is stable and $F(i) = 0$. $F(i)$ can be described as equation (3), where T_1 is a threshold.

$$F(i) = \begin{cases} 1 & \text{if } \forall k \in (i, i-1, \dots, i-L+1) : |G(k) - A(i)| \geq T_1 \\ 0 & \text{Otherwise} \end{cases} \quad (3)$$

3) *Peak Values Extraction*: The algorithm in the node automatically extracts the sequence of peak values, i.e., local maxima and minima from a continuous fluctuation signal which produces a successive sequence of " $F(i) = 1$ ". Using an array $P = \{(k, t_k, p_k), 1 \leq k \leq N\}$ to restore the peak values, where k is No., t_k is the sample time, p_k is the amplitude. Then filter out peaks with small amplitude or short duration, delete these elements in the array P if their del_flag equal 1. The del_flag is defined in equation (4), where T_2 and T_3 are thresholds. So only the amplitude and duration of a peak is large enough will be adopted. In the Fig. 3, there are six peak values (including the initial and terminal values of the signal), which are denoted by cycles.

$$del_flag(k) = \begin{cases} 1 & \text{if } p_k - p_{k-1} < T_2 \text{ or } t_k - t_{k-1} < T_3 \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

4) *Feature Variables Computing*: From the array $P = \{(k, t_k, p_k), 1 \leq k \leq N\}$ of peak values, the algorithm can compute feature variables of the continuous fluctuation signal.

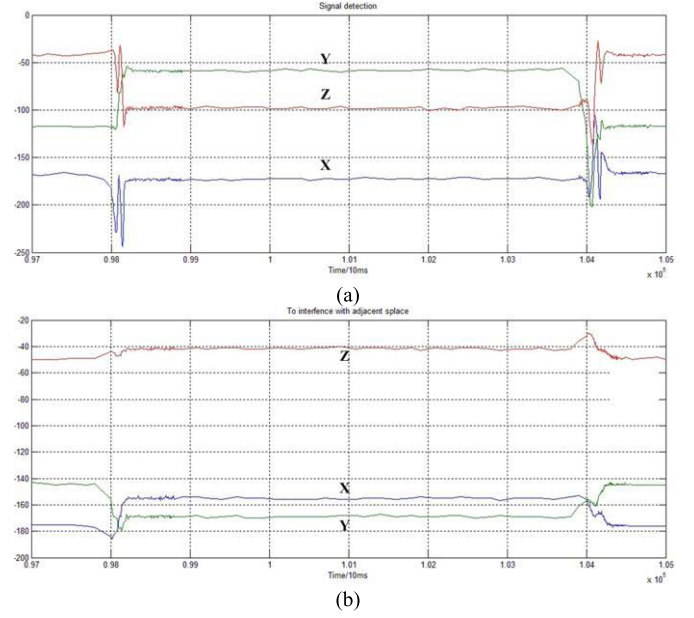


Fig. 4. The parking signal and interference signal of a vehicle. (a) Parking signal. (b) Interference signal.

The variable *Peak_counter* indicates the number of the magnetic signal fluctuations. The minimum tracks the local minimums in a continuous fluctuation signal while the maximum tracks the local maximums. The variable *Max_amplitude* is the amount of change between the minimums and maximum. The variable *Drift* indicates the change before and after the signal fluctuations, as shown in Fig. 3.

$$Drift = abs(flat_previous - flat_current). \quad (5)$$

B. Classification Criteria

1) *Parking Detection*: In our experiments, we found that some of the vehicles disturb the geomagnetic field greater. They often bring interference to the sensors at the adjacent parking spaces. For simplicity, we define two kinds of signals: 1) The parking signal which is perceived by the sensor node deployed on the vehicle parked space. 2) The interference signal which is perceived by the sensors at the neighbor parking spaces. The parking signal and the interference signal of a vehicle are shown in Fig. 4.

Through extensive data analysis shows that, under normal circumstances, the interference waveform's change is relatively flat, although there will be the formation of stable drift, but no occurrence of the waveform several times undulating. Interference brought by the by-pass moving vehicles on the road produces a short period of time fluctuations, and thus will not form a stable drift.

At present, the existing research results are mostly based on the stable drift of parking stop phase to detect vehicle occupancy, but the interference of neighbor parking will also form similar drift, especially in the case of low signal-to-noise ratio, as shown in Fig. 4. A vehicle presence detection method is proposed based on the integrated magnetic signal characteristics of a vehicle's entering and parking stop phases. According to our statistics, almost all magnetic signals of

TABLE I
PRE-CLASSIFIED SAMPLES

w_1 : Parking signal				w_2 : Interference signal			
No.	v_1	v_2	v_3	No.	v_1	v_2	v_3
1	5	100	50	1	1	20	15
2	6	80	30	2	0	10	8
...	...	...	...	...	...	...	...

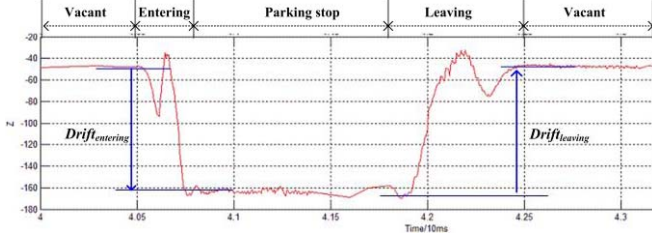


Fig. 5. V_1 equals V_2 approximately.

vehicle parking own two features: frequent fluctuation and a stable drift, this is what we proposed algorithm based on where. Abstract the vehicle signal for the three dimension data vector model, $x = (v_1, v_2, v_3)^T$, where x denotes the classified objects, v_1, v_2, v_3 are the three feature variables *Peak_counter*, *Max_amplitude* and *Drift*, respectively. v_1 and v_2 are the characteristics of the vehicle entering phase, v_3 represents the characteristic of the parking stop phase.

Given a linear discriminant function

$$W(x) = w^T x + w_0 \quad (6)$$

As shown in Table I, the goal is to learn the weights w using a set of n labeled samples x_i (i.e., each x_i has a class label w_1 or w_2). A two-category classifier with a discriminant function of the form $g(x)$ uses the following rule: Decide w_1 if $g(x) \geq 0$ and w_2 if $g(x) < 0$.

2) *Leaving Detection*: Leaving can be regarded as an inverse process of parking. In the case of no interference, the leaving detection simplified as the magnetic signal recovered to be same as the background earth magnetic field. But in the reality applications, there has overlap interference. Usually when the vehicle left, the magnetic signal is not recovered to the background value. Therefore, we choose the vehicle entering fluctuation signal as the referential signal. Leaving detection is based on computing the similarity between the entering and leaving signal.

- 1) Give $B(i)$ is the background magnetic value, $A(i)$ is the current magnetic signal value. If $R(i) = 1$, it can decide the magnetic $A(i)$ is approximately equal to $B(i)$.

$$R(i) = \begin{cases} 1 & \text{if } |A(i) - B(i)| \leq T_4 \\ 0 & \text{Otherwise} \end{cases} \quad (7)$$

- 2) As shown in Fig. 5, when the vehicle is completely stopped, a stable drift to the geomagnetic produced. We can get the drift through the calculation of difference between the state of vacant and occupied, denoted by $Drift_{entering}$. From Fig. 5 we can see, an opposite variation emerged when the vehicle left,

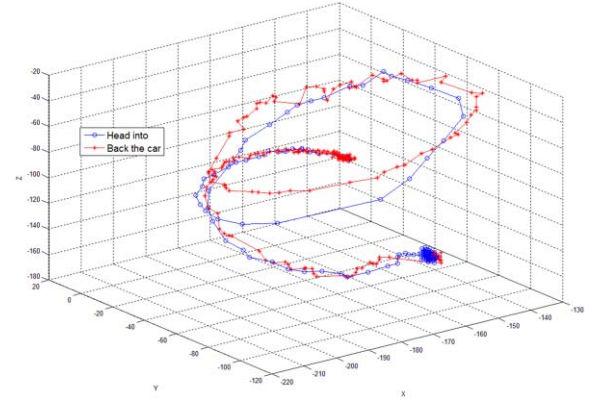


Fig. 6. The magnetic signal of vehicle entering and leaving has symmetry, the asterisk is entering signal and the circle is leaving signal.

denoted by $Drift_{leaving}$. $Drift_{entering}$ equals $Drift_{leaving}$ approximately.

$$\begin{aligned} dist(Drift_{entering}, Drift_{leaving}) &= \frac{abs(Drift_{entering1} - Drift_{leaving1})}{\max(Drift_{entering1}, Drift_{leaving1})} \end{aligned} \quad (8)$$

$$drift_sim = \begin{cases} 1 & \text{if } dist(Drift_{entering}, Drift_{leaving}) \in [0, C_1] \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

Using $dist$ function to compute the similarity, a decision criterion uses the following rule: $Drift_{entering}$ and $Drift_{leaving}$ are similar if $dist(Drift_{entering}, Drift_{leaving}) \in [0, C_1]$, where C_1 is a given constant, in our experiment, it is 0.1.

- 3) We also note that the magnetic signal of vehicles entering and leaving have some symmetry. Fig. 1 presents an even symmetry. This symmetry is more obvious in Fig. 6.

P is an array includes peak values extract from the entering fluctuation. Peak points in the array are stored in chronological order. Then invert the array P and renumbered, and the final result is $P_{entering} = \{(i, t_i, x_i), 1 \leq i \leq N\}$. Array $P_{leaving}$ is the peak point collection of the leaving phase magnetic fluctuations. $P_{leaving} = \{(j, s_j, y_j), 1 \leq j \leq M\}$, elements are stored in chronological order. The distance between $P_{entering}$ and $P_{leaving}$ can be computed by function $S(P_{entering}, P_{leaving})$. A decision criterion uses the following rule: it can be considered to the correspond waveform of $P_{entering}$ and $P_{leaving}$ are symmetrical if $S(P_{entering}, P_{leaving}) \in [0, C_2]$, where C_2 is a given constant.

$$S(P_{entering}, P_{leaving}) = \sum_{i=1}^M \frac{abs(y_i - x_i)}{\max(y_i, x_i)} / M \quad (10)$$

$$fluct_sim = \begin{cases} 1 & \text{if } S(P_{entering}, P_{leaving}) \in [0, C_2] \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

A two-category classifier with a discriminant function of the equation Leaving uses the following rule:

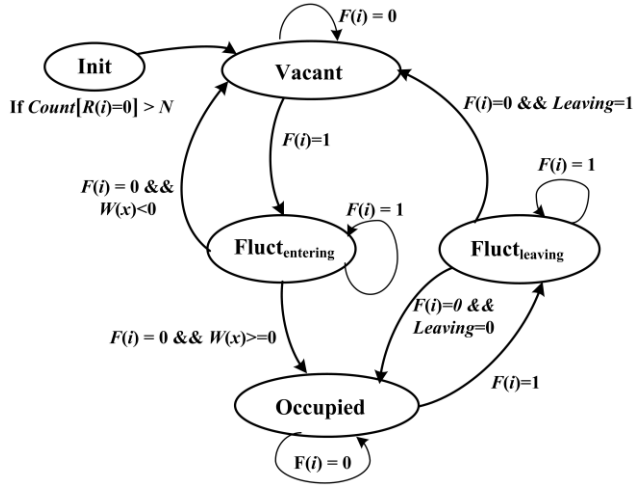


Fig. 7. The state machine of the proposed algorithm.

Given the parking space is Occupied, decide w_1 (vacant) if $Leaving = 1$ and w_2 (occupied) if $Leaving = 0$.

$$Leaving = \begin{cases} 1 & \text{if } (drift_sim = 1 \ \&\& \ R(i) = 1) \\ & \text{or } (drift_sim = 1 \ \&\& \ fluct_sim = 1) \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

C. Detection Algorithm

Occupancy of parking space is determined by a state machine, as shown in Fig. 7. The state machine consists of:

State: {Init, Vacant, Fluct_{entering}, Occupied, Fluct_{leaving}}

Input $[F(i), W(x), Leaving]$: {0, 1}

Output $[D(i)]$: {0, 1}

The Boolean flags $F(i)$ is passed to the state machine. Their main objective is to trace the fluctuate signals that are caused by vehicles entering or leaving. $D(i)$ is the output of the detection state machine. Only when the state machine is in “Occupied” or “Fluct_{leaving}”, $D(i) = 1$, otherwise, $D(i) = 0$. The following section is a walk-through of the state machine’s logical flow.

S1 “Init”: Assuming there is no vehicle in the parking space when the sensor node is being deployed. It will go into state S1 and start initializing the background value with the environmental magnetic field.

S2 “Vacant”: After a pre-defined initializing time, it will jump to the state S2 where the arrays $P_{entering}$ and $P_{leaving}$ are initialized. It will jump to state S3 there has an “ $F(i) = 1$ ” and trace the possible entering fluctuation.

S3 “Fluct_{entering}”: It was found that a vehicle entering the parking space produces a successive sequence of “ $F(i) = 1$ ” and this state is used to track such a sequence. The algorithm extracts the peak values of the fluctuation and restores them in the array $P_{entering}$. If there is any “0” reading from $F(i)$, the algorithm computes $W(x)$. If $W(x) \geq 0$, it will immediately jump to state S4. Otherwise, $W(x) < 0$, it will jump back to state S2.

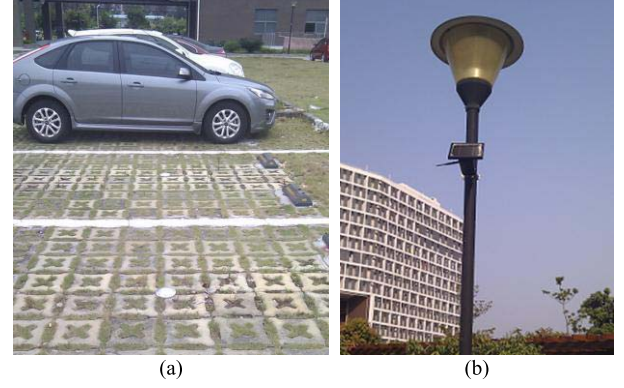


Fig. 8. Devices are deployed in roadside parking spaces. (a) Sensor nodes. (b) Router.

S4 “Occupied”: Staying in this state implies the vehicle is on the sensor node, and the parking space is occupied. It initializes the array $P_{leaving}$. It will jump to S5 when the sensor detected a possible leaving fluctuation.

S5 “Fluct_{leaving}”: Within this state, it extracts the peak values of the fluctuation and restores them in the array $P_{leaving}$. If there is any “0” reading from $F(i)$, the algorithm computes the function $Leaving$. It will jump back to state S4 if $Leaving = 0$. Otherwise, $Leaving = 1$, it will jump to state S2.

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. System Introduction

The sensor node has Zigbee [28] based communication modules and it consists of HMC5883L [29] magnetic sensor. The Zigbee wireless module is using the CC2530 MCU [30]. For battery powered sensor nodes, the lifetime is a critical factor. The sensor node has no specific responsibility for maintaining the network infrastructure because it is an end-device [28]. So it doesn’t exchange packet for network maintaining after joining the network. Sensor node samples the magnetic signal periodically. When it detected a car arrival or leave, it transmits one binary detection flag to base station. And it goes to sleep when no task to save energy. Using a pair of 2500mAh AA Li batteries parallel connected, the sensor node can continuously work for 5 years without changing battery.

To test the proposed system, we deploy eighty-two sensor nodes in the roadside parking. As shown in Fig. 8 (a). Devices with AMR sensors are nailed in the center of the parking spaces. The routers which are equipped a solar panel for forwarding parking message are fixed on the roadside light, as shown in Fig. 8 (b).

We developed a server system using Java language and MySQL database. As shown in Fig. 9, using the graphical client interface, users can know which parking space is vacant or occupied, and the occupied duration of each parking space. As shown in Fig. 9(a), the nodes which ID signed as integer, such as 6, 3, 5, and so on, are router nodes. The nodes are magnetic sensor nodes, which ID started with “B”, such as B27, B26, and so on. The sensor node to be colored as a green

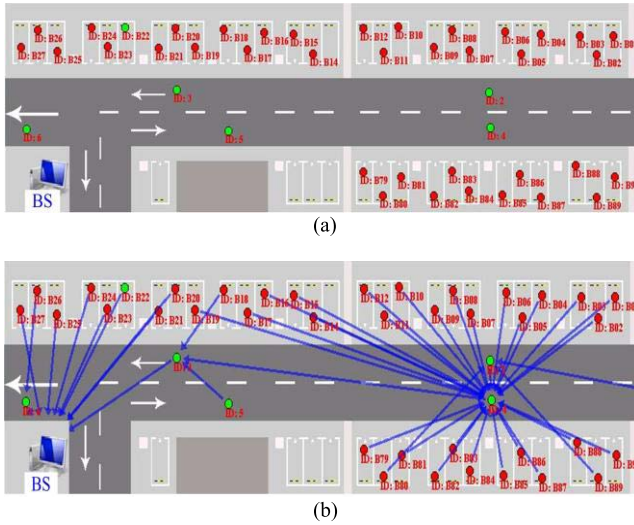


Fig. 9. Pictures of the management system. (a) Sensor nodes deployed in the parking spaces. (b) Topology of the network.

TABLE II
EXPERIMENT PARAMETERS

Parameter	Description	Value
L	Smooth buffer length	30
T_1	A threshold for $F(i)$	10
T_2	An amplitude threshold for del_flag	20
T_3	A duration threshold for del_flag	3ms
T_4	A threshold for $R(i)$	5
C_1	A threshold for $drift_sim$	0.1
C_2	A threshold for $fluct_sim$	0.2

dot indicates the according parking space is vacant, on the other hand, a red dot represents the parking space is occupied. In Fig. 9 (b), these lines with arrow describe the wireless network topology.

B. Experimental Results

In our experiments, parameters related to the parking detection algorithm are given in Table II. Fig. 10 shows the detection results of a parking space B19 for successive 8 days. X, Y, Z are the raw magnetic field, and their value please refer to the right coordinate axis. $D(i)$ is the detection result whose value please refer to the left coordinate axis, where value 0 indicates the parking is “vacant”, and 1 stands for “occupied”. As shown in Fig. 11, it describes the daily occupancy time of the parking space B19 for successive 8 days.

In order to verify the proposed Parking Occupancy Detection Algorithm (PODA for short), we deployed 82 nodes in the parking spaces. The system has operated for six months. According to the large number of measured data, we analyze the performance of the PODA using the detection accuracy of vehicle parking and leaving. Table III shows the average accuracy of vehicle parking detection of PODA is about 98.9%.

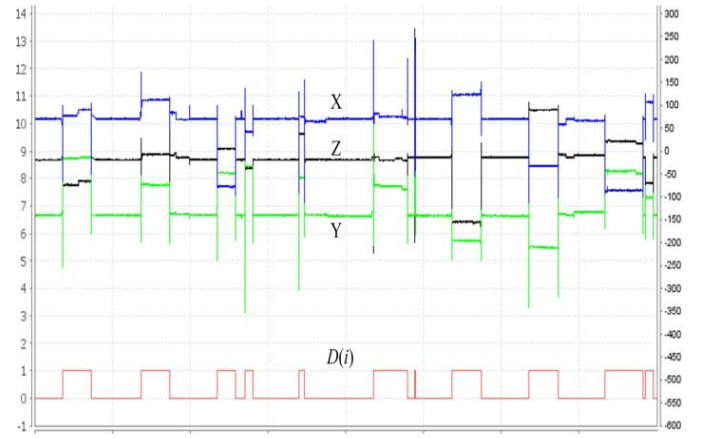


Fig. 10. The detection results of a parking space for successive 8 days.

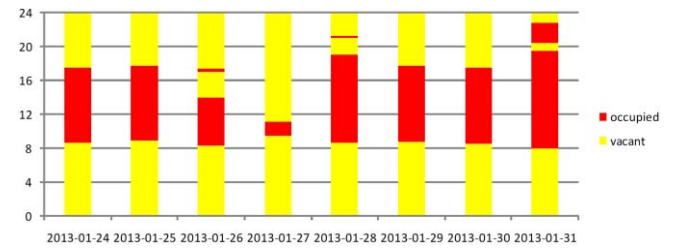


Fig. 11. The occupied time of a parking space.

TABLE III
VEHICLE PARKING DETECTION RESULT

The number of times of vehicle parking	Detected the number of vehicle parking of PODA	Accuracy of PODA
32896	32535	0.989

We are considered the cases in which a car has two or more times attempts before it comes to a parking stop, and also considered the similar situation in the car leaving phase. Assume D_1 is the magnetic disturbance duration of the first attempt, and D_3 is the duration of the second attempt. Between D_1 and D_3 , there has D_2 which is the duration of a car beyond the sensor’s sensing range.

First case, $D_2 > 0$, during D_2 , the magnetic signal recovered to be same as the background magnetic field. The algorithm will ignore the magnetic signals of D_1 , and recognize D_3 as a new entering phase.

Second case, $D_2 = 0$, i.e. between two attempts, the car is within the sensor’s sensing range, as shown in Fig. 12. Our algorithm will combine the signal of D_1 and D_3 to be a car entering signal because it adopts a delay aware fluctuation tracking mechanism. Vehicle detection is based on peak values of the fluctuation wave, so that this case will not affect the performance of the algorithm, but will result in a larger delay of the vehicle detection.

C. Comparative Analysis

The prior work [24] proposed An Adaptive Threshold Detection Algorithm (ATDA for short). Using the magnetic

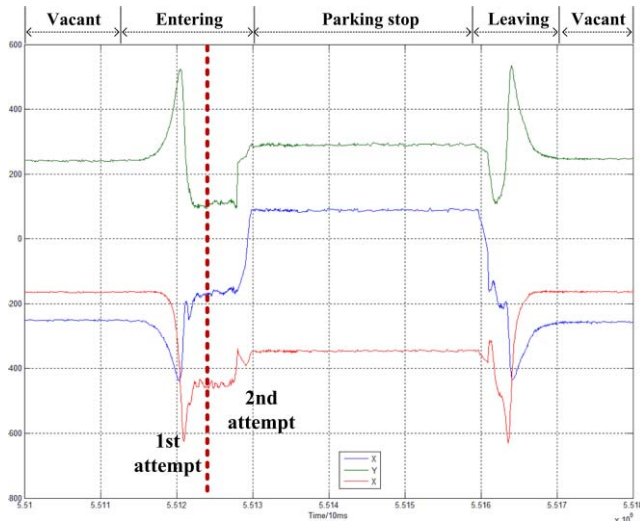


Fig. 12. The magnetic signature of a car parking after two attempts.

TABLE IV
THE CONTRAST OF VEHICLE PARKING DETECTION ACCURACY
BETWEEN PODA AND ATDA

Case	Situation	The number of vehicle parking	Accuracy of PODA	Accuracy of ATDA
1	No vehicle on both left and right	720	0.998	0.96
2	A vehicle on the left	720	0.99	0.95
3	A vehicle on the Right	720	0.99	0.95
4	Two vehicles on both left and right	720	0.989	0.94

signal collected from sensor nodes, we run PODA and ATDA on PC, respectively. Table IV shows the test results of vehicle parking. The experiment results show that in the first case two algorithms have best performance. In the case 2-4, interference signal caused by vehicles on neighbor parking spaces has a negative impact effect on the detection performance. And PODA is more accurate for vehicle detection than ATDA. ATDA smoothes out the entering fluctuation of the magnetic signal, and it is only considers the signal's characteristic of the parking stop phase. As shown in Fig. 13–14, there doesn't have a threshold to distinguish the characteristic between parking stop and interference. In the low SNR (signal-to-noise ratio) situation, the frequent up-and-down fluctuation of the magnetic signal is a key characteristic for the vehicle parking detection. With the feature is designed, PODA has a better performance than ATDA.

In the parking charge application, it is billed by the duration of occupancy. For the parking management company, cost to customers' overcharged is more expensive than less-charged. So the vehicle leaving detection requires higher accurate than the parking detection. Vehicle leaving detection results is shown in Table V. The Accuracy of PODA is better than

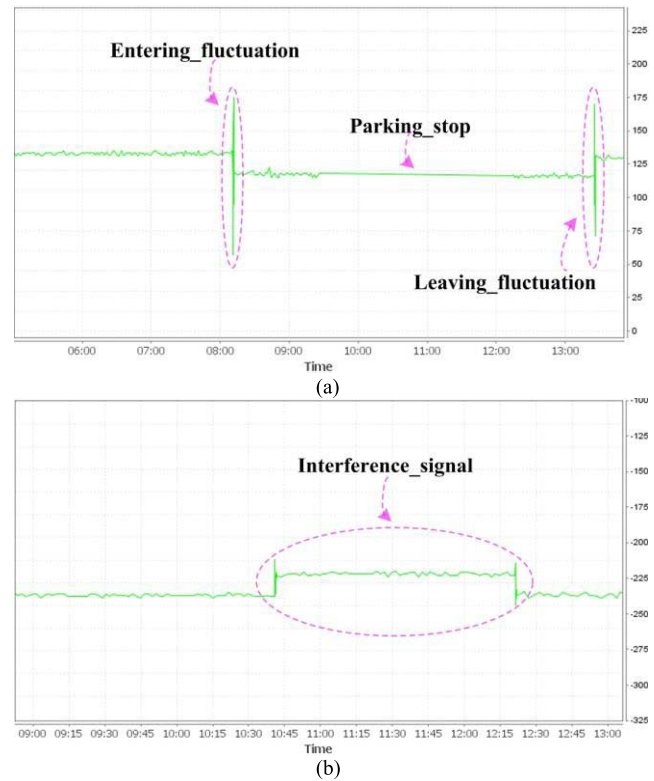


Fig. 13. The signal comparison of parking and interference. (a) The signal of a vehicle parking. (b) The signal of interference.



Fig. 14. The signal comparison of parking and interference.

TABLE V
DETECTION RESULTS OF THE VEHICLE LEAVING

The number of times of the vehicle leaving the parking space	Accuracy of PODA	Accuracy of ATDA
32896	0.99	0.96

ATDA. In ATDA, the leaving detection is simplified as the magnetic signal recovered to be same as the background earth magnetic field. But in the roadside parking applications, there has overlap interference. Usually when the vehicle left, the magnetic signal is not recovered to the background value. As shown in Fig. 15, after the parking space is occupied, there has interference signal overlap on the parking signal. When the vehicle left, the magnetic signal did not recover to the background value. It is an obvious drift compared to

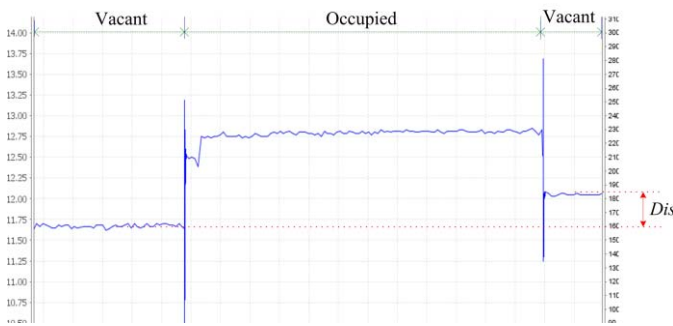


Fig. 15. After the parking space is occupied, there has interference signal overlap on the parking signal. When the vehicle left, the magnetic signal did not recover to the background value.

the background magnetic field between two vacant statuses. Using ATDA, the algorithm cannot do a correct vehicle leaving detection.

IV. CONCLUSION

In this paper, a parking occupancy detection algorithm based on AMR sensor is presented. It focuses on characteristic of the interference signal caused by adjacent vehicles, and proposes a more accurate detection algorithm than existing algorithms both on the detections of vehicle parking and leaving. The parking occupancy monitoring system, which includes eighty-two sensor nodes, was deployed on a roadside parking. The system has been running reliably for six months. Extensively experiment results show that the system detection accuracy is better than 98%.

REFERENCES

- [1] Q.-J. Kong, Z. Li, Y. Chen, and Y. Liu, "An approach to urban traffic state estimation by fusing multisource information," *IEEE Trans. Intell. Transp. Syst.*, vol. 10, no. 3, pp. 499–511, Sep. 2009.
- [2] G. Alessandretti, A. Broggi, and P. Cerri, "Vehicle and guard rail detection using radar and vision data fusion," *IEEE Trans. Intell. Transp. Syst.*, vol. 8, no. 1, pp. 95–105, Mar. 2007.
- [3] P. N. Pathirana, A. E. K. Lim, A. V. Savkin, and P. D. Hodgson, "Robust video/ultrasonic fusion-based estimation for automotive applications," *IEEE Trans. Veh. Technol.*, vol. 56, no. 4, pp. 1631–1639, Jul. 2007.
- [4] V. Cevher, R. Chellappa, and J. H. McClellan, "Vehicle speed estimation using acoustic wave patterns," *IEEE Trans. Signal Process.*, vol. 57, no. 1, pp. 30–47, Jan. 2009.
- [5] B. Guo, M. S. Nixon, and T. Damarla, "Improving acoustic vehicle classification by information fusion," *Pattern Anal. Appl.*, vol. 15, no. 1, pp. 29–43, Feb. 2012.
- [6] C. N. Dickson, A. M. Wallace, M. Kitchin, and B. Connor, "Improving infrared vehicle detection with polarisation," in *Proc. IET Intell. Signal Process. Conf. (ISP)*, Dec. 2013, pp. 1–6.
- [7] M. Chen and T. Chang, "A parking guidance and information system based on wireless sensor network," in *Proc. IEEE Int. Conf. Inf. Autom. (ICIA)*, Jun. 2011, pp. 601–605.
- [8] Q. Zhou, G. Tong, B. Li, and X. Yuan, "A practicable method for ferromagnetic object moving direction identification," *IEEE Trans. Magn.*, vol. 48, no. 8, pp. 2340–2345, Jul. 2012.
- [9] J. Petrou, P. Skafidas, E. Hristoforou, "Electronic toll and road traffic monitoring system using 3-D field AMR sensors," *Sensor Lett.*, vol. 11, no. 1, pp. 91–95, 2013.
- [10] B. Koszteczyk and G. Simon, "Magnetic-based vehicle detection with sensor networks," in *Proc. IEEE Int. Instrum. Meas. Technol. Conf.*, May 2013, pp. 265–270.
- [11] J. Pelegrí Sebastián, J. A. Lluch, and J. R. L. Vizcaino, "Signal conditioning for GMR magnetic sensors: Applied to traffic speed monitoring GMR sensors," *Sens. Actuators A, Phys.*, vol. 137, no. 2, pp. 230–235, 2007.

- [12] S. Y. Cheung, S. Coleri, B. Dundar, S. Ganesh, C.-W. Tan, and P. Varaiya, "Traffic measurement and vehicle classification with a single magnetic sensor," California PATH, Univ. California, Berkeley, Berkeley, CA, USA, Tech. Rep. UCB-ITS-PWP-2004-7, 2004.
- [13] S.-Y. Cheung and P. P. Varaiya, "Traffic surveillance by wireless sensor networks: Final report," California PATH, Univ. California, Berkeley, Berkeley, CA, USA, Tech. Rep. UCB-ITS-PRR-2007-4, 2007.
- [14] M. H. Kang, B. W. Choi, K. C. Koh, J. H. Lee, and G. T. Park, "Experimental study of a vehicle detector with an AMR sensor," *Sens. Actuators A, Phys.*, vol. 118, no. 2, pp. 278–284, 2005.
- [15] R. Wang, L. Zhang, R. Sun, J. Gong, and L. Cui, "EasiTia: A pervasive traffic information acquisition system based on wireless sensor networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 12, no. 2, pp. 615–621, Jun. 2011.
- [16] L. Zhang, R. Wang, and L. Cui, "Real-time traffic monitoring with magnetic sensor networks," *J. Inf. Sci. Eng.*, vol. 27, no. 4, pp. 1473–1486, 2011.
- [17] J. Zhu, C. Hongbing, S. Jie, and L. Haitao, "Data fusion for magnetic sensor based on fuzzy logic theory," in *Proc. Int. Conf. Intell. Comput. Technol. Autom. (ICICTA)*, Mar. 2011, pp. 87–92.
- [18] S. Taghvaeeyan and R. Rajamani, "The development of vehicle position estimation algorithms based on the use of AMR sensors," *IEEE Trans. Intell. Transp. Syst.*, vol. 13, no. 4, pp. 1845–1854, Dec. 2012.
- [19] G. Yao, J. Wang, X. Ran, Z. Li, and L. Yao, "Topology study on parking guidance and information system based on CAN," in *Proc. Int. Conf. Comput., Mechatron., Control, Electron. Eng. (CMCE)*, Aug. 2010, pp. 420–422.
- [20] P. Baronti, P. Pillai, V. W. C. Chook, S. Chessa, A. Gotta, and Y. F. Hu, "Wireless sensor networks: A survey on the state of the art and the 802.15.4 and ZigBee standards," *Comput. Commun.*, vol. 30, no. 7, pp. 1655–1695, May 2007.
- [21] S. Lee, D. Yoon, and A. Ghosh, "Intelligent parking lot application using wireless sensor networks," in *Proc. Int. Symp. Collaborative Technol. Syst.*, May 2008, pp. 48–57.
- [22] S. Coleri, S. Y. Cheung, and P. Varaiya, "Sensor networks for monitoring traffic," in *Proc. Allerton Conf. Commun., Control, Comput.*, 2004, pp. 32–40.
- [23] A. Haoui, R. Kavalier, and P. Varaiya, "Wireless magnetic sensors for traffic surveillance," *Transp. Res. C, Emerg. Technol.*, vol. 16, no. 3, pp. 294–306, 2008.
- [24] S. Y. Cheung, "Traffic surveillance by wireless sensor networks," Ph.D. dissertation, Univ. California, Berkeley, Berkeley, CA, USA, 2006.
- [25] J. Ding, S.-Y. Cheung, C.-W. Tan, and P. Varaiya, "Signal processing of sensor node data for vehicle detection," in *Proc. 7th Int. IEEE Conf. Intell. Transp. Syst.*, Oct. 2004, pp. 70–75.
- [26] J. Gu, Z. Zhang, F. Yu, and Q. Liu, "Design and implementation of a street parking system using wireless sensor networks," in *Proc. 10th IEEE Int. Conf. Sens. Ind. Informat.*, Jul. 2012, pp. 1212–1217.
- [27] (Aug. 2014). *SENSIT*. [Online]. Available: <http://www.nedapavi.com/products/sensit/>
- [28] ZigBee Alliance. (Aug. 2014). *ZigBee PRO Specification*. [Online]. Available: <http://www.zigbee.org>
- [29] (Aug. 2014). *Three-Axis Magnetic Sensor HMC5883L*. [Online]. Available: http://www51.honeywell.com/aero/common/documents/myaerospacecatalog-documents/Defense_Brochures-documents/HMC5883L_3-Axis_Digital_Compass_IC.pdf
- [30] (Aug. 2014). *CC2530*. [Online]. Available: <http://www.ti.com/cn/product/cn/cc2530>



Zusheng Zhang received the B.S. degree from the Hunan University of Science and Technology, Xiangtan, China, in 2003, the M.S. degree from Nankai University, Tianjin, China, in 2006, and the Ph.D. degree from the Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China, in 2010. He is currently an Assistant Researcher with the Dongguan University of Technology, Dongguan, China. His research interests include intelligent transportation system and wireless sensor networks.



Ming Tao received the B.S. degree from Anhui University, Hefei, China, in 2007, and the M.S. and Ph.D. degrees from the South China University of Technology, Guangzhou, China, in 2009 and 2012, respectively. He is currently an Assistant Researcher with the School of Computer, Dongguan University of Technology, Dongguan, China. His research interests include the next-generation wireless/mobile networks.



Huaqiang Yuan was born in 1966. He received the Ph.D. degree from Shanghai Jiao Tong University, Shanghai, China, in 1996. He is currently a Professor with the School of Computer, Dongguan University of Technology, Dongguan, China. His primary research interests include signal processing and data mining.